

Evaluating the Faithfulness of Saliency-Based Explanations in Reinforcement Learning: Do They Reflect What Actually Drives an Agent's Decision-Making?

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Abstract

This project investigates whether saliency-based explanations of a reinforcement learning (RL) agent's decisions are faithful to the actual factors driving its behavior, or whether they merely create an illusion of interpretability. A Deep Q-Network (DQN) agent was trained to navigate a grid-world environment containing obstacles, achieving an 86% success rate across 50 evaluation episodes. Faithfulness was evaluated through a targeted perturbation method, in which the obstacle identified as most salient (via input-gradient saliency) was removed from the observation and the agent's resulting action was compared to its original decision, against two control conditions — removing the least-salient obstacle and removing a randomly chosen obstacle. This evaluation was conducted in three successive phases of increasing scale and rigor: an initial 30-trial pilot on a single trained agent, a 150-trial follow-up on the same agent, and a final 300-trial study across three independently trained agents (100 trials each). The pilot suggested no relationship between saliency and causal importance (top-salient and control faithfulness both 20.0%; Spearman $\rho = 0.136$, $p = 0.202$). The 150-trial run on the same agent revealed a weak but statistically significant positive correlation ($\rho = 0.182$, $p = 0.0001$). The final multi-seed study confirmed and strengthened this finding: pooled across three agents, top-salient faithfulness (29.7%, 95% CI [24.8%, 35.1%]) was significantly higher than both the least-salient control (16.3%, $p = 0.00013$) and the random-obstacle baseline (19.0%, $p = 0.00012$), while the two control conditions did not differ from each other ($p = 0.699$) — confirming that the effect is specifically attributable to saliency rank, not merely to obstacle removal in general. The positive correlation between saliency and causal effect was independently significant in each of the three agents ($\rho = 0.135$ – 0.277 , all $p < 0.02$), demonstrating that this weak-but-real signal is reproducible across independently trained models rather than an artifact of a single agent. Overall, saliency-based explanations for this agent carry genuine, non-random, but quantitatively weak information about causal importance — a nuanced

conclusion made possible only by explicitly tracking how the finding evolved as statistical power increased across the three phases of this study.

1. Introduction

Reinforcement learning agents are increasingly proposed for real-world applications, including autonomous navigation and robotics. However, these agents typically operate as “black boxes” — their internal decision-making process is not directly interpretable to human observers. This lack of transparency raises concerns about trust, particularly in safety-critical applications.

Explainable AI (XAI) methods, such as saliency maps, attempt to address this by highlighting which parts of an agent’s input most strongly influenced its decision. However, an important open question is whether these explanations are faithful — that is, whether they accurately reflect the actual causal factors behind a decision, or whether they merely produce a plausible-looking but misleading visualization.

This project investigates that question directly: do saliency-based explanations of an RL agent’s navigation decisions truthfully reflect what drives those decisions, or do they primarily create an illusion of interpretability? A secondary, methodological aim of this project is to demonstrate — using the project’s own experimental history as a case study — how conclusions about explanation faithfulness can change, and become more trustworthy, as the sample size and experimental controls are scaled up in stages.

2. Related Work

Gradient-based saliency was originally proposed for image classifiers as a way to visualize which pixels most influence a network’s output (Simonyan, Vedaldi, & Zisserman, 2013), and has since been widely adapted to other domains, including reinforcement learning, where perturbation- and gradient-based saliency methods have been used to visualize what Atari-playing agents attend to during gameplay (Greydanus, Koul, Dodge, & Fern, 2018). However, a separate line of work has raised doubts about whether such visualizations are trustworthy: Adebayo, Gilmer, Muelly, Goodfellow, Hardt, and Kim (2018) showed that several popular saliency methods for image classifiers produce nearly identical maps even when model weights or training labels are randomized — a “sanity check” failure suggesting some saliency methods are more sensitive to input structure than to the model’s actual learned parameters. More broadly, Rudin (2019) has argued that post-hoc explanation methods for black-box models should not be trusted for high-stakes decisions, and that interpretable-by-design models are preferable wherever feasible. This project extends this line of skepticism specifically to gradient-based saliency in the RL setting, using a causal perturbation test rather than a purely qualitative or visual assessment, and — unlike much of

the sanity-check literature, which typically reports a single pass/fail test on a single model — explicitly evaluates how such a test’s conclusions change with sample size and across independently trained models, using a continuous rank-correlation measure alongside matched control conditions.

3. Methodology

3.1 Environment

A custom grid-world environment was implemented using the Gymnasium API. The environment consists of a 5×5 grid containing a fixed start position, a fixed goal position, and three randomly placed obstacles per episode. The agent can move up, down, left, or right; movement into an obstacle or the grid boundary is blocked. The agent receives a reward of +10 for reaching the goal, −5 for colliding with an obstacle, and a small shaped reward based on the change in Euclidean distance to the goal on all other steps, encouraging efficient navigation.

The observation space encodes the grid as three separate one-hot layers (agent position, goal position, obstacle positions), avoiding a naive integer encoding that would introduce a false ordinal relationship between categorical values.

3.2 Agents

A Deep Q-Network (DQN) agent was trained using Stable-Baselines3, with a multi-layer perceptron policy (two hidden layers of 128 units) over 150,000 timesteps, achieving an 86% success rate across 50 evaluation episodes. This original agent was used for the first two experimental phases (Sections 3.5.1–3.5.2). For the final phase (Section 3.5.3), three additional agents were independently trained from different random seeds (seed = 0, 1, 2), using identical architecture and hyperparameters, to test whether findings generalized beyond a single trained model. In total, four independently trained agents contributed data across this study.

3.3 Saliency Map Generation

For a given decision, a saliency map was computed by taking the gradient of the selected action’s Q-value with respect to each element of the input observation. The resulting gradient magnitudes were reshaped into three spatial layers corresponding to the agent, goal, and obstacle channels, providing a visual indication of which input regions most strongly influenced the agent’s decision.

3.4 Perturbation Test

To test the faithfulness of these explanations, a targeted perturbation method was applied:

for each trial, the obstacle cell with the highest saliency value was identified and removed from the observation (set to zero), while all other inputs remained unchanged. The modified observation was then passed back through the trained policy, and the resulting action was compared to the agent’s original action. Two control conditions were used for comparison: removing the least-salient obstacle, and removing a randomly selected obstacle (introduced in Phase 3; see Section 3.5.3). To obtain a continuous measure of faithfulness beyond this binary top-1 test, all three obstacles in each trial were individually removed and the resulting action recorded, allowing a Spearman rank correlation to be computed between each obstacle’s saliency value and whether removing it actually changed the agent’s decision.

3.5 Experimental Progression

This study was conducted in three successive phases, each motivated by limitations identified in the previous one. Reporting all three phases, rather than only the final result, is a deliberate methodological choice: it demonstrates concretely how an under-powered study can produce a misleading conclusion, and how that conclusion changes as the study is scaled up.

3.5.1 Phase 1 — Pilot (n = 30, single agent). An initial 30-trial pilot applied the top-salient vs. least-salient perturbation test and the rank-correlation measure to the original trained agent.

3.5.2 Phase 2 — Single-agent scale-up (n = 150, same agent). To address the small sample size of the pilot, the same procedure was repeated with 150 trials on the same agent, along with a Q-value margin analysis (splitting trials by the gap between the top and second-best action) to test whether faithfulness depended on decision confidence.

3.5.3 Phase 3 — Multi-seed replication (n = 300, three agents). To address the remaining concern that Phase 1–2 results might reflect properties of one specific trained agent rather than a general property of gradient-based saliency in this environment, three additional agents were trained from different random seeds, and 100 trials were run per agent (300 trials total, 900 individual obstacle removals). A third control condition — removing a randomly selected obstacle — was added at this stage to distinguish “saliency carries no information” (top-salient rate $\approx$ random-baseline rate) from “saliency carries some information beyond simply removing any obstacle” (top-salient rate $>$ both controls, controls similar to each other).

3.6 Experimental Pipeline and Design Choices

Figure 1 summarizes the full evaluation pipeline used in every trial across all three phases, from observation to aggregated statistics.

Figure 1. The experimental pipeline applied per trial. Each observation produces a saliency

map, from which the top-, least-, and (in Phase 3) a randomly selected obstacle are chosen for removal; the resulting change in policy output is recorded and aggregated across trials into the statistics reported in Section 4.

Several methodological choices underlie this pipeline; Table 1 summarizes the reasoning behind each.

Table 1. Key design choices and their rationale.

Design choice	Rationale
DQN as the RL algorithm	A simple, well-established value-based baseline with a direct Q-value output, avoiding the added complexity of computing saliency for a stochastic policy (as in PPO or A2C)
Custom 5×5 grid-world	Provides full experimental control over obstacle placement, action space, and reward structure, isolating the faithfulness question from confounds present in higher-dimensional environments
One-hot layered observation	Avoids a false ordinal relationship between agent, goal, and obstacle positions that a naive integer encoding would introduce
Input-gradient saliency	The original and still most widely used gradient-based saliency method, providing a natural baseline against which more recent methods (e.g., SHAP, Integrated Gradients) could later be compared
Perturbation-based faithfulness test	Directly tests a causal claim (does removing this obstacle change the decision?) rather than relying on a purely visual or qualitative judgment of the saliency map
Least-salient and random controls	Distinguishes “saliency carries no information” from “saliency carries some information beyond simply removing any obstacle” — a distinction a single top-salient condition alone cannot make
Spearman rank correlation	Appropriate for relating an unbounded, non-normally distributed saliency value to a binary outcome (action changed / did not change), without assuming a linear relationship
Multi-seed replication (Phase 3)	Tests whether findings reflect a general property of the trained policy class rather than an artifact of one specific trained instance

4. Results

4.1 Phase 1: Pilot (n = 30)

Removing the most-salient obstacle changed the agent's action in 6/30 trials (20.0%); removing the least-salient obstacle produced an identical rate (6/30, 20.0%). The Spearman correlation between saliency and causal effect across all 90 individual obstacle removals was weak and not statistically significant ($\rho = 0.136$, $p = 0.202$). At this stage, the data were consistent with the interpretation that saliency carried no detectable signal about causal importance.

4.2 Phase 2: Single-agent scale-up (n = 150)

With five times the data on the same agent, top-salient faithfulness rose to 54/150 (36.0%, 95% CI [28.8%, 43.9%]), compared to 36/150 (24.0%, 95% CI [17.9%, 31.4%]) for the least-salient control — a 12-percentage-point gap (paired comparison, $p \approx 0.065$, trending but not conventionally significant). Critically, the rank correlation across all 450 obstacle removals was now statistically significant: $\rho = 0.182$, $p = 0.0001$. A Q-margin split (median = 0.130) found no difference in faithfulness between high-margin (37.3%) and low-margin (34.7%) decisions at this sample size.

4.3 Phase 3: Multi-seed replication (n = 300, three agents)

Condition	Action changed	Rate	95% CI (Wilson)
Most-salient obstacle removed	89/300	29.7%	[24.8%, 35.1%]
Least-salient obstacle removed (control)	49/300	16.3%	[12.6%, 20.9%]
Random obstacle removed (baseline)	57/300	19.0%	[15.0%, 23.8%]

The top-salient condition was significantly higher than both the least-salient control (paired comparison: 69 trials where only top-salient changed the action vs. 29 where only the control did, $p = 0.00013$) and the random baseline (47 vs. 15, $p = 0.00012$). Critically, the two control conditions did not differ significantly from each other (32 vs. 24, $p = 0.699$) — the result expected if both represent “non-informative” obstacle choices, confirming that the elevated top-salient rate is specifically attributable to the saliency ranking rather than to obstacle removal in general.

4.4 Reproducibility across independently trained agents

Agent (seed)	Top-salient	Least-salient	Random	Spearman ρ	p-value
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Seed 0	38.0%	19.0%	20.0%	0.135	0.019
Seed 1	29.0%	19.0%	22.0%	0.189	0.001
Seed 2	22.0%	11.0%	15.0%	0.277	<0.0001
Pooled (n=300)	29.7%	16.3%	19.0%	0.202	<0.000001

The rank correlation between saliency and causal effect was positive and statistically significant in all three independently trained agents, with the top-salient condition exceeding both controls within every individual agent as well as in the pooled data. The magnitude of the effect varied across agents (p ranging from 0.135 to 0.277), but its direction and statistical significance were consistent — directly addressing the concern that a single-agent result might not generalize.

4.5 Decision confidence (Q-value margin): a reversal on larger data

With the full 300-trial dataset, trials were again split at the median Q-value margin (0.157). Unlike the null result observed in Phase 2 ($n = 150$, single agent), the larger multi-seed sample revealed a statistically significant difference: low-margin (near-tied) decisions were faithful 36.7% of the time (55/150), compared to 22.7% (34/150) for high-margin (confident) decisions (two-proportion z-test, $z = 2.65$, $p = 0.008$). This is discussed in Section 5.3.

4.6 Action distribution across agents

Agent (seed)	Action 0	Action 1	Action 2	Action 3
Seed 0	29	0	0	71
Seed 1	34	2	0	64
Seed 2	74	1	0	25

All three independently trained agents show a strong skew toward two of the four possible actions, with the near-complete absence of actions 1 and 2 in every agent — but which of the two dominant actions prevails differs by agent (seeds 0 and 1 favor action 3; seed 2 favors action 0). This indicates that the direction of the policy’s bias is arbitrary and seed-dependent, while the fact of a strong two-action skew is a consistent, reproducible property of the trained policy across agents.

5. Discussion

5.1 What the three-phase progression demonstrates

The central methodological finding of this project is visible only by comparing all three phases together (summarized below), and is arguably as important as the substantive finding about saliency itself:

Metric	Phase 1 (n=30)	Phase 2 (n=150)	Phase 3 (n=300, 3 agents)
Top-salient faithfulness	20.0%	36.0%	29.7%
Best available control	20.0% (least)	24.0% (least)	16.3–19.0% (least/random)
Gap (top – best control)	0.0 pp	12.0 pp	10.7–13.4 pp
Spearman ρ	0.136	0.182	0.202 (pooled)
p p-value	0.202 (n.s.)	0.0001	<0.000001

The pilot’s apparent null result was a direct consequence of insufficient statistical power: an effect size as small as $\rho \approx 0.15$ – 0.20 requires on the order of 100+ paired observations to detect reliably, and 90 observations (Phase 1) fell short of that threshold. Rather than being a contradictory or unstable result, Phase 1 is best understood as an underpowered measurement of the same underlying effect later confirmed in Phases 2 and 3 — the point estimate of ρ increased only modestly ($0.136 \rightarrow 0.182 \rightarrow 0.202$) across phases, while the precision of that estimate (and thus its statistical significance) improved substantially. This project’s own history is a direct illustration of why a single small-sample faithfulness study — of the kind common in student and early-career XAI projects — should not be treated as a final answer. Put plainly: this project’s own experimental history illustrates the danger of underpowered XAI evaluations — a pilot study can produce a confident-looking null result that a larger, better-controlled follow-up subsequently overturns, not because the underlying phenomenon changed, but because the pilot was never large enough to see it.

5.2 Interpretation of the final multi-seed findings

The multi-seed study (Phase 3) provides the most rigorous evidence in this project: gradient-based saliency carries a real, statistically robust, and cross-agent-reproducible signal about causal importance, but one that is quantitatively weak. The three-way comparison is central to this interpretation — the fact that the least-salient and random-obstacle controls did not differ from each other (16.3% vs. 19.0%, $p = 0.699$), while the top-salient condition significantly exceeded both, rules out the possibility that any single-

obstacle removal simply tends to change the agent’s action regardless of which obstacle is chosen. The elevated rate is specifically tied to the saliency ranking.

At the same time, a rank correlation of $\rho \approx 0.20$ means saliency rank explains only about 4% of the variance in whether an obstacle removal actually changes the agent’s decision. A human relying on the saliency map to predict the agent’s causal sensitivity would do reliably better than chance, but would still be wrong the majority of the time for any single decision.

5.3 Reconciling the Q-margin reversal

The finding that low-margin (near-tied) decisions were more faithful than high-margin decisions (36.7% vs. 22.7%, $p = 0.008$) did not appear in the smaller Phase 2 sample and only emerged with the full 300-trial dataset. This is consistent with the effect being real but modest in size — similar in character to the overall saliency correlation, it required a larger sample to detect reliably. Substantively, the result supports the “holistic decision-making” hypothesis raised earlier in this project’s analysis: when the agent’s top two candidate actions have similar Q-values, a relatively small change to the input (removing a single obstacle) is more likely to tip the balance toward the second-best action. When one action’s Q-value dominates by a wide margin, removing a single obstacle is less likely to change the ranking, regardless of how salient that obstacle appeared. This offers a plausible mechanistic account for why saliency-based faithfulness is weak rather than absent: saliency captures something real about the agent’s sensitivity to an input region, but that sensitivity is gated by how close the agent’s decision was to begin with — a factor gradient-based saliency, computed only with respect to the chosen action’s Q-value, does not directly represent.

5.4 Threats to Validity

Internal validity. Only one saliency method (input gradients) was evaluated. Gradient-based saliency is known in the broader XAI literature to sometimes behave differently from other explanation families (e.g., perturbation-based or Shapley-value methods), so the specific magnitude of the correlation reported here ($\rho \approx 0.20$) should be understood as a property of this method on this agent, not of “saliency maps” as a general category. The Q-margin analysis (Section 4.5) and the action-distribution analysis (Section 4.6) were exploratory rather than pre-registered, and should be treated as hypothesis-generating rather than confirmatory.

External validity. All experiments were conducted in a single custom 5×5 grid-world with a four-action discrete space and a single RL algorithm (DQN). The weak effect size observed here may not generalize to more complex, higher-dimensional environments, such as the pixel-based Atari domains studied by Greydanus et al. (2018), or to policy-gradient algorithms such as PPO, where the relationship between saliency and causal importance could be stronger, weaker, or qualitatively different. The multi-seed design (Phase 3)

resolves the concern that results reflect one specific trained instance, but it does not resolve generalization across environments, saliency methods, or algorithm families.

Construct validity. This study uses “the agent’s action changed after obstacle removal” as an operational proxy for “this obstacle was causally important to the decision.” This proxy is imperfect: an obstacle could be genuinely relevant to the agent’s reasoning without being the single deciding factor in a four-way action choice, and conversely, an action change could occur for reasons unrelated to the removed obstacle’s importance (e.g., the removal altering the shaped-distance component of the observation indirectly). The rank-correlation measure partially mitigates this by using a continuous rather than binary signal, but the underlying causal claim being tested is necessarily narrower than “does the agent understand or attend to this obstacle” in any richer cognitive sense.

5.5 Future Work

The most immediate extension would be to test whether the weak-but-real signal reported here ($\rho \approx 0.20$) is specific to gradient-based saliency or characteristic of causal faithfulness testing more broadly. This would involve repeating the same perturbation-and-correlation protocol with alternative explanation methods — Shapley-value-based approaches (SHAP) and Integrated Gradients are natural next candidates, since both are widely used and, unlike input gradients, are designed to satisfy specific axiomatic faithfulness properties by construction.

A second direction would be to test generalization across the RL algorithm family, extending the multi-seed methodology used here (Phase 3) to a policy-gradient method such as PPO, to determine whether value-based and policy-based agents exhibit similar saliency–causality relationships.

A third direction would be to scale the environment itself: moving from the current 5×5 discrete grid to larger grids, continuous control tasks, or vision-based observations (e.g., Atari-style pixel inputs) would test whether the weak effect size observed here is a property of small, low-dimensional action spaces specifically, or persists in higher-dimensional settings where saliency maps are more commonly deployed in practice.

Finally, the Q-margin reversal reported in Section 4.5 — that faithfulness was higher for near-tied decisions than for confident ones — was an exploratory, post hoc finding. A pre-registered follow-up study designed specifically to test this “decision-margin” hypothesis, ideally across multiple environments, would be needed to establish it as a general property rather than a pattern specific to this dataset.

5.6 Implications

These findings offer a more nuanced conclusion than a simple “saliency works” or “saliency

doesn't work" verdict, and this nuance was only reachable by tracking the finding's evolution across three phases of increasing rigor. For applications such as autonomous robotics, where understanding why an agent made a decision can be critical for safety and accountability, a weak-but-real, cross-agent-reproducible correlation is arguably a more dangerous outcome than either a strong correlation or a clean null result: it is strong enough to look convincing and produce plausible-seeming explanations, consistent enough across models to survive a naive "try it on another seed" check, but too weak to be trusted for any individual high-stakes decision. This supports the broader XAI concern — echoed by both Adebayo et al. (2018) and Rudin (2019) — that post-hoc explanation methods warrant independent, quantitative faithfulness testing conducted at adequate statistical power before being deployed in high-stakes settings. Within the experimental setting of this study specifically, saliency maps were insufficiently reliable to serve as a stand-alone explanation for any individual decision — a qualitative assessment, or even a single small-scale quantitative test, would not have revealed the gap between their apparent and actual reliability documented here.

6. Conclusion

Across three successive experimental phases — a 30-trial pilot, a 150-trial single-agent follow-up, and a final 300-trial study across three independently trained agents — this project found that gradient-based saliency maps for a DQN navigation agent carry a real, statistically significant, and cross-agent-reproducible but quantitatively weak relationship with the agent's true causal decision-making (pooled Spearman $\rho = 0.202$, $p < 0.000001$; individual-agent ρ ranging from 0.135 to 0.277, all significant). The most-salient obstacle changed the agent's action significantly more often than either a least-salient or a randomly selected control obstacle, while the two controls did not differ from each other — isolating the effect specifically to the saliency ranking. This conclusion was reachable only because the study did not stop at its initial 30-trial pilot, which had suggested no relationship at all; scaling the sample size and adding independent training seeds revealed a consistent, if modest, signal that a single small-scale test would have missed entirely. The overall picture is one of weak, partial, but genuine and reproducible faithfulness: saliency maps for this class of agent are not meaningless, but they remain far too unreliable to serve as a trustworthy explanation for any single decision — underscoring both the substantive need for empirical faithfulness validation before deploying explainable AI in safety-critical robotic systems, and the methodological need to conduct such validation at a sample size and replication scope sufficient to distinguish a real weak effect from statistical noise.

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Tools & Libraries Used

- Google Colab — development environment
- Python 3
- NumPy — array operations and grid representation
- Gymnasium — RL environment API
- Stable-Baselines3 — DQN implementation and training
- PyTorch — underlying neural network framework (via Stable-Baselines3)
- Matplotlib — visualization of saliency maps and results